

Instructor Notes — Submitting AI Jobs to the Anvil Supercomputer

Beginner Track 1 - AI for All - Sept 15 2026 - 1:00-2:30 PM CT

Pre-session checklist (T-minus — do these BEFORE the workshop)

Accounts (the #1 success factor): - [] Every attendee has an ACCESS ID (access-ci.org) and it's added to the workshop allocation on Anvil — confirm the allocation + user list with Preston Smith / RCAC (rcac-help@purdue.edu) at least 2 weeks out - [] Confirm the account type: ACCESS credit transfer vs. NAIRR Pilot allocation (NAIRR = automatic on approval, no transfer step — affects the “getting your own allocation” slide wording) - [] SSH keys: attendees without keys use the **OnDemand browser path** (ondemand.anvil.rcac.purdue.edu -> set up SSH key from OnDemand) — put this URL on slide 10 and in the handout; do NOT budget lab time for key generation - [] Get the Anvil username format right in the handout: `x-<ACCESS username>` (not the ACCESS username itself)

Lab staging: - [] Stage labs/ on Anvil in a world-readable directory (e.g. `/anvil/projects/<workshop-project>/ai-works`) and a short URL/QR code on slide 10 and 18 - [] On the Anvil login node, run and CAPTURE (screenshot for slides): - `showpartitions` — verify `debug` and `gpu-debug` have idle capacity at 1 PM CT on a recent weekday (if `gpu-debug` idle is 0 at 1 PM historically, plan the front-of-room demo fallback) - `sfeatures` — `g[000-015] A100 / h[000-020] H100` node names - `module spider pytorch` — **pin the exact PyTorch module or container for Lab 4** (train-gpu.sbatch has TODO comments at both lines; fill in the verified one) - `module load conda; python3 -c "import sklearn; print(sklearn.__version__)"` — verify Lab 3's import works in the default conda module; if not, capture the venv/pip fallback commands - [] Dry-run ALL FOUR labs on Anvil itself (submit each `.sbatch`, check outputs) — this catches partition, account, and module problems while there's still time to fix them - [] Verify the workshop allocation's `-A` account string exactly (`mybalance` on the login node) — pre-fill it in the staged copies of the `.sbatch` files so attendees' first submission works without editing (leave `<ACCOUNT>` in the handout copies as a teaching moment)

Room/logistics: - [] Projector + your own SSH session pre-logged-in and font-size bumped (16pt+) - [] Wi-Fi for attendees; know the room's fallback if SSH is blocked (OnDemand browser path) - [] Helper-to-attendee ratio ~1:10; brief helpers on the troubleshooting table in handout.md - [] Attendees arrive at 1:00 from lunch — start with the why (slides 2–3), not housekeeping

Timing plan (90 min)

Clock	Segment	Watch for
1:00–1:15	Framing, slides 1–9	Don't overrun slide 4 (partition table) — it's reference, not lecture
1:15–1:27	Lab 1: connect, mybalance, showpartitions	SSH-key people eat the budget — route them to OnDemand immediately
1:27–1:39	Lab 2: first sbatch	Checkpoint: everyone's <code>slurm-*.out</code> exists. This is the never-cut lab
1:39–1:57	Lab 3: CPU AI job (digits)	If >5 min behind at start, convert to front-of-room demo + they submit-and-check-later
1:57–2:12	Lab 4: GPU job (PyTorch)	The wow moment (nvidia-smi shows A100). Protect 10 min for it
2:12–2:15	Stretch: job array	Only if on schedule; fast finishers only

Clock	Segment	Watch for
2:15–2:30	Wrap-up, slides 15–18, Q&A	Getting-your-own-allocation slide is the #1 question — have allocations.access-ci.org and nairrpilot.org ready

Fallbacks (decide thresholds BEFORE the session)

Situation	Fallback
Attendee can't SSH (no key, blocked port)	OnDemand browser: ondemand.anvil.rcac.purdue.edu -> Clusters -> Shell access — everything except the stretch lab works there
debug queue full at 1 PM	Scripts run unchanged on shared (the default partition) — <code>-p debug -> -p shared</code> ; say it once, everyone moves
gpu-debug saturated	Instructor demos Lab 4 from front; attendees submit and get a PENDING job + “check your output after the session” — frame as the real HPC experience
sklearn/torch missing from default conda module	pip-in-venv fallback (capture the exact commands pre-session); or switch Lab 4 to the NGC container line (already in train-gpu.sbatch)
Running >10 min behind at 2:00	Cut Lab 3's “try different cpus-per-task” extension and the stretch; NEVER cut Lab 2 or the allocation slide

Known pitfalls (from verified RCAC docs)

- **Missing -A account is the #1 first-submission error** (“failed to map user” / “invalid account specified”) — it's in the handout troubleshooting table and slide 8 says it twice.
- **Anvil username is not ACCESS username**: it's `x-<ACCESSname>`. The handout says this at the SSH step.
- **No partition specified -> shared (default)**: fine for Lab 3's script but say it explicitly — people assume “no `-p` = no queue”.
- **Password auth is rejected on Anvil** — SSH keys or OnDemand only.
- **gpu-debug limits: 30 minutes, 2 GPUs, 1 running job per user** — train-gpu.sbatch asks `-t 00:25:00` and `--gres=gpu:1` to stay safely inside.
- **debug limits: 2 nodes, 2 hrs, 1 running job per user** — fine for all CPU labs; attendees who resubmit before their first finishes will sit PENDING (that's a teaching moment for `squeue`'s reason column, not an error).
- **Charging**: node-exclusive partitions (wholenode/wide) bill 128 cores even if you request 1 — one slide-17 bullet, prevents confused `mybalance` questions later.
- **GRES on GPU nodes**: `sfeatures` shows `gpu:4` per g/h node — request `--gres=gpu:1`, not “a GPU node” in the script name.

Lab files map (what's in labs/)

File	Lab	Verifies
hello.py + hello.sbatch	2	syntax-checked locally; runs on plain python3
train_digits.py + train-digits-cpu.sbatch	3	executed locally in a venv: 0.989 test accuracy, metrics JSON written
train_gpu.py + train-gpu.sbatch	4	syntax-checked; requires torch + GPU node — dry-run on Anvil pre-session (see checklist)
train_sweep.py + array.sbatch	stretch	executed locally with SLURM_ARRAY_TASK_ID=2: works, writes sweep_02.json

After the session

- Collect `sacct -j <jobid>` outputs from anyone whose GPU job didn't finish in time
- Share the lab repo URL + handout PDF with the closing session (4:20 PM) crowd
- Debrief with Preston Smith: gpu-debug capacity at session time, whether a dedicated workshop queue is possible next year